

The Etymology of Albanian *dorë*

PIE **de-r-* “hand, open palm,” the **ǵh^hesr-* side reading, and the ARM → HAND grammaticalization pathway

Companion paper to *The Hand That Gives* (Paper 1 of the *der-* project)

Working paper v2 — for scholarly review

J.M. Chavis and Claude (Anthropic AI)

ABSTRACT

Albanian *dorë* “hand” (consonantal plural *duar*) has been derived in two competing ways since the nineteenth century. The **der-* side reading, originating with Gustav Meyer’s *Albanesische Studien I* (1883), groups *dorë* with the published PIE entry **der-* “Spanne der Hand, ausgebreitete Hand, Handfläche” (Pokorny IEW 203). The lineage runs through La Piana (1937, 1942), Belardi (1948), Pokorny (1959), Schindler (1967), and Frisk (1960–72). The **ǵh^hesr-* side reading consolidated through Demiraj (1990, 1997) and was inherited by Orel (1998), Vermeer (2008), and Beekes (2010); it groups *dorë* with the **ǵh^hesr-* “hand” entry at Pokorny IEW pp. 447–448, alongside Greek χείρ, Hittite *kessar*, Armenian *jeṛn*, and Tocharian A *tsar* / B *ṣar*.

This paper develops the case for the **der-* side reading on three grounds.

Phonological: **der-* → *dorë* requires only sound laws all parties accept — Szemerényi’s Law on the consonant-stem r-stem nominative, regular Albanian **ē* → *o*, the **eh₂-* feminine extension, unstressed final **ā* → *-ë*. Every **ǵh^hesr-* side reconstruction must accommodate a step **ǵh^h → d* in Albanian, where the regular reflex of PIE **ǵh^h* is *dh*. **Historiographic:** the post-1990 reading rests, in its foundational citation (Demiraj 1997: 140 → Tagliavini *L’albanese di Dalmazia* p. 101), on a page whose content does not in fact address Albanian *dorë*. **Comparative:** the post-2000 specialist reconstruction (Vermeer 2008) has independently moved to **ǵh^hēr-* — a sibilant-less form differing from **dēr-* by one phoneme — and Vermeer himself describes the sound law on which Huld’s exclusion of La Piana’s **dōrom* rested as having no reliably agreed example.

A concluding section (§8) addresses the deeper question of where PIE **de-r-* itself comes from. The cross-linguistically well-documented ARM → HAND grammaticalization (Heine and Kuteva 2002) is supported by a published Indo-European parallel (Matasóvič 2009: 103–104, on Proto-Celtic **dowsant-* “arm, hand” from PIE **dóws-* “arm, fore-arm”), by Sanskrit lexicographic evidence (the s-stem arm-noun *doṣ-/doṣán-* with the rich compound stem *dor-* and its 18-inch metrological reading), and by a within-Sanskrit attestation (Pārvatī “taking up her son with both her hands (*dos*)” in the *Śivapuraṇa* 2.4.18).

The morphological frame is supplied by three convergent published treatments of the PIE **-r* element: Lipp 2009 on the body-part r/n heteroclite class, Bauhaus 2019 on PIE **-r* as a locative case-ending, and Kahl 2024 on the locative origin of the comparative suffix **(t)ero-*. A formal constraint qualifies the deeper hypothesis: the well-attested IE arm-root has o/u-vocalism (**dows-*), not e-vocalism (**der-*), so the deeper unification requires a Pre-PIE compositional analysis developed in the broader four-paper project. Probability: **de-r-* → *dorë* etymology, 55–70%; deeper ARM → HAND hypothesis, 15–25%.

1. Introduction

Albanian *dorë* “hand” (consonantal plural *duar*) is the standard Albanian word for “hand.” Its etymology has been contested for more than a century, and the contest is not settled even within the assumptions of its currently dominant reconstruction.

Two PIE sources are on offer. The first derives *dorë* from the **ǵh^hesr-* “hand” entry at Pokorny IEW pp. 447–448, the same root that produces Greek χείρ, Hittite *kessar*, Armenian *jeṛn*, and Tocharian A *tsar* / B *šar*. This is the post-1990 standard, canonicalized by Demiraj (1990, 1997) and inherited by Orel (1998: 99), Vermeer (2008), and Beekes (2010). It requires positing a sound change **ǵh^h → d* in Albanian, where the regular reflex of PIE **ǵh^h* is *dh*. Within the **ǵh^hesr-* side reading itself, three accounts disagree on how to derive the long vowel: Demiraj’s compensatory lengthening (from loss of **-s-*), Huld’s analogical lengthened grade, and Vermeer’s reconstruction without sibilant (**ǵh^hēr-*, with the source of length left open).

The second derives *dorë* from the **der-* “Spanne der Hand” entry at Pokorny IEW p. 203, with the reconstruction written here as PIE **de-r-* “hand, open palm.” This reading originates with Gustav Meyer’s *Albanesische Studien I* (1883), runs through La Piana (1937, 1942), Belardi (1948), Pokorny (1959), Schindler (1967), and Frisk (1960–72, *GEW* I 430–431), and was a standard handbook treatment for half a century before Demiraj 1990. The **der-* side reading uses only regular Albanian sound changes.

This paper develops the case for the **der-* side reading. §2 surveys the two parallel lineages. §3 examines Pokorny IEW p. 447 directly, where Pokorny’s own text rules out **ǵh^hesr- → dorë* on phonological grounds. §4 examines the Tagliavini citation that grounds the 1990 consolidation. §5 examines the supporting pillars cited by Orel: Hamp 1966 and Huld 1984. §6 examines Vermeer 2008. §7 develops the reconstruction **der-s → *děr → *děr-eh₂- → *děr-ā → dorë* in detail. §8 turns to the deeper question of the ARM → HAND grammaticalization pathway, the Matasóvič Proto-Celtic parallel, the Sanskrit s-stem arm-noun, the morphological frame of Lipp/Bauhaus/Kahl, and the formal constraint that qualifies the deeper

hypothesis. The paper makes no claim about which specific Indo-European forms do or do not belong to the **der-* entry of IEW 203; decisions about the cognate set are reserved for the companion papers.

2. Two Lineages

The **der-*side reading for *dorë* begins with Gustav Meyer's *Albanesische Studien I* (1883). The **ġh^hesr-*side reading has antecedents in Bopp (1854) and consolidated in Meyer's own 1891 *Etymologisches Wörterbuch der albanesischen Sprache*, where Meyer reversed his earlier position. The two readings coexisted in the published literature for nearly seven decades.

La Piana's monograph (1937) and short *Indogermanische Forschungen* note (1942) reactivated Meyer's 1883 attribution after the **ġh^hesr-*side had been the working hypothesis since 1891. Belardi's 1948 "Ittito *ki-eš-šar*, greco *χείρ*" provided the decisive phonological argument against the **ġh^hesr-* derivation, developed in §3 below. Pokorny's IEW (1959) placed *dorë* under **der-* at IEW 203 — citing La Piana — while also entering it under the **ġh^hesr-* entry at IEW 447 with explicit acknowledgment of the phonological problem. Schindler (1967), Frisk's *Griechisches Etymologisches Wörterbuch* (1960–72, I 430–431), and the *Diccionario Griego-Español* preserve the **der-*side reading.

Demiraj's "Sull'etimologia di alb. *dorë*" (*Indogermanische Forschungen* 95, 1990) and the corresponding entry in *Albanische Etymologien* (1997) consolidated the **ġh^hesr-*side reading for the post-1990 specialist literature. Orel's *Albanian Etymological Dictionary* (1998: 99) adopted **ġh^hesr-* as the standard for the entry. Vermeer (2008) worked within the Demiraj/Huld frame. Beekes (2010) placed *dorë* primarily in the *χείρ* entry (pp. 1620–1) and, in the *δῶρον* 2 entry (p. 363), cited La Piana 1942 only to refer the reader to Demiraj 1997. The post-1990 consensus is unanimous for the **ġh^hesr-*side reading. The task of this paper is to demonstrate that the consensus rests on the 1990 consolidation addressed in §4, rather than on independent phonological grounds.

3. Pokorny IEW p. 447: Doubt at the Source

Demiraj's *Albanische Etymologien* (1997: 140) cites Pokorny IEW p. 447 as the position the new analysis overturns. Direct examination of that page reveals that Pokorny's own text already ruled out **ġh^hesr-* → *dorë* on phonological grounds. The relevant passage from the entry "1. *ġhesor-*, *ġhesr-* 'Hand'?":

*Da arm. jern auch auf *ġher-ϰ zurückgehen könnte und ebenso die gr. Formen besser aus *ġher-s erklärt werden können, da ferner alb. dorë f. 'Hand', Pl. konson. duar eine Grundform *ġhēr- erfordert (*ġhēs- hätte *dostrë ergeben), wäre mit Belardi (Riv. Studi Orient. 23, 69 ff.) zu erwägen, ob nicht fürs Hitt. und Toch. Metathese von *-rs- zu *-sr- anzunehmen sei, und die Gruppe 1 von 2 zu trennen und zu *ġher- 'greifen' zu stellen sei.*

Three features of this entry deserve notice.

The question mark on the gloss. Pokorny titles the entry “**ĝhesor-*, *ĝhesr-* ‘Hand’?” — with a question mark. The meaning of the etymon is flagged as uncertain; handbook compilers use this signal when the etymon may not actually carry the meaning the conventional treatment assigns to it.

The Belardi argument in Pokorny’s own voice. Translated: “Albanian *dorë* f. ‘hand’, consonantal plural *duar*, requires a base form **ĝhēr-* (**ĝhēs-* would have yielded **dostrë*).” This is Belardi 1948 in Pokorny’s voice: **ĝh^hesr-* cannot have produced Albanian *dorë* because the regular Albanian outcome of **ĝh^hēs-* would have been **dostrë*. Pokorny’s tentative alternative is **ĝher-* “greifen” (IEW pp. 442–443), whose semantic field is enclosed places: Greek *χόρτος* “enclosed place,” *χορός* “dancing-place,” Latin *cohors*, *hortus*. The body-part word produced by this root in Greek is *χόριον* “afterbirth” (the membrane that encloses the fetus), not *χείρ*. Pokorny’s reassignment moves *χείρ* to a root with no other “hand” reflex to motivate the move — and offers no escape from the same Albanian sound-law problem, since **ĝh^h → d* would be no more regular for **ĝher-* than for **ĝh^hesr-*.

The unsolved problem. The Albanian sound-law difficulty is left unsolved by either of Pokorny’s two reconstructions on IEW p. 447. The **de-r-* reading (PIE **d* giving Albanian *d* by regular sound change) is the only proposal among the published candidates that does not require positing the irregular **ĝh^h → d* shift in Albanian.

4. The 1990 Reading and the Tagliavini Reference

Demiraj’s *Albanische Etymologien* (1997: 140) consolidates the **ĝh^hesr-* side reading in three steps. (1) Demiraj acknowledges that Pokorny doubted the **ĝh^hesr-* → *dorë* reconstruction. (2) Demiraj acknowledges that Pokorny preferred the **der-* side reading — Meyer’s connection of *dorë* to Greek *δῶρον* “palm of hand,” with La Piana in agreement, the entry sitting on Pokorny IEW p. 203. (3) Demiraj overturns Pokorny’s preference with a single citation: Tagliavini *L’albanese di Dalmazia*, p. 101 — “Diese Deutung ist mit Tagliavini (AD 101) auszuschließen” (“This interpretation is to be excluded with Tagliavini (AD 101)”). The substance of the consolidation therefore rests on what Tagliavini *L’albanese di Dalmazia* page 101 actually contains.

L’albanese di Dalmazia (Tagliavini 1937) is a study of the Albanian dialect of Borgo Erizzo near Zara in Dalmatia; its bulk is an alphabetical glossary. Page 101 falls within that glossary, in the “d-” entries. The page contains entries for *dëkum* “morto,” *dëlme* “pecora,” *dër* (preposition) “fino, mentre,” *dër* (-a) “porta,” *dërdh* (verb) “versare, spandere,” and *dërdhum* “diarrea, erpete.” It does not contain a discussion of Albanian *dorë* “hand.” It does not mention Greek *δῶρον*. It does not contain a rejection of any etymology applicable to *dorë*. Tagliavini’s actual *dorë*

entry is on AD p. 106, five pages later; its six lines cite Meyer, Brugmann, Walde-Pokorny, Pedersen, and Thumb, and do not engage the **der-/δῶρον* alternative as something to be rejected.

The page reference at Demiraj 1997: 140 is the kind of slip that can occur in any wide-ranging etymological work. The substantive point is not who made the mistake but that the citation does not deliver what subsequent literature has taken it to deliver. Orel's *Albanian Etymological Dictionary* (1998: 99) repeats "TAGLIAVINI Dalmazia 101" in its bibliography for the *dorë* entry, carrying the reference forward into the consolidated apparatus. Pokorny's own text at IEW 447 had stated, in his own voice, that **ĝh^hesr-* could not produce *dorë*; the consolidation effectively substitutes an external authority for Pokorny's own phonological judgment, and the external authority does not on inspection make the relevant statement.

Tagliavini's own published positions bear on the substance. His 1936 inaugural lecture *La lingua albanese (Studi Albanesi V-VI: 5-33)* states at p. 13 that Albanian responds to PIE **ĝ/ĝh^h* with *dh*, with *d* only "in certain cases" with no rule specifying the conditioning; and at p. 14 that **ē → o* is regular in Albanian (example: *lodh(ë)* "tired" < **lēd-*). Tagliavini's own published statements on Albanian historical phonology thus document that the **ĝh^h → d* step required by the **ĝh^hesr-* side view is irregular, and that the **ē → o* step required by the **der-* side reading is regular.

5. Hamp 1966 and Huld 1984: The Supporting Pillars

Orel (1998: 99) cites two pre-2000 sources as supporting the **ĝh^hesr- → dorë* derivation: Hamp 1966 and Huld 1984. Both deserve direct examination.

5.1 Hamp 1966

Eric Hamp's "The Position of Albanian" (1966: 115) treats *dorë* in six lines, in a section explicitly framed as review and correction of Porzig's *Gliederung*, not original research. The language throughout is tentative: "are taken back to" (passive, no agent), "is something like," "(fem. in Albanian < old neut pl. ?)" with a question mark, "the apparent reconstructions," "an alternative." Hamp proposes a reconstruction **ĝhē(s)ră* for the singular and does not engage Pokorny IEW p. 203, Meyer, La Piana, or Belardi. Huld (1984: 54) reports the passage as "Hamp (1966a:115) noted that the singular arose from a collective neuter plural, *ĝhēsṛā*" — dropping the question mark and converting the tentative speculation into a noted finding.

A further observation bears on §7 below: Hamp's reconstruction **ĝhē(s)ră* uses the same morphological move available to a **dēr-eh₂-* reconstruction — an athematic stem reanalyzed as a feminine collective, with the lengthened root vowel as part of the inherited form. The **der-* side reading does not depart from Hamp's morphological pattern; it differs from Hamp on the initial consonant and on the

source of the long **ē*.

5.2 Huld 1984

Martin Huld's *Basic Albanian Etymologies* (1984: 54) is the more substantive pillar. His objection is that La Piana's specific reconstruction — **dōrom*, with **ō* — is defective: Albanian *o* does not come from PIE **ō*. Huld concludes that Pokorny's preferred derivation "was rightly dismissed by Tagliavini."

This conclusion runs into two issues. First, it depends on the AD 101 citation addressed in §4: Tagliavini AD 101 does not dismiss the etymology in question. Second, Huld's phonological objection applies to La Piana's specific form **dōrom*, but the **der*-side reconstruction developed in §7 is not **dōrom*. The reconstruction is **dēr-eh₂-* — an **ē*-grade form, not an **ō*-grade form. Per Huld's own statement on Albanian phonology, *o* is regularly from **ē*. The chain **dēr-eh₂-* → **dēṛā* → *dorë* uses precisely the sound law Huld endorses. Huld's filter, applied to the **ē*-grade reconstruction, passes it. Huld himself includes IEW 203 *√der*- "Spanne der Hand" in his terminal bibliographic references for the entry.

6. Vermeer 2008: The Standard Reading Without the Sibilant

Vermeer's "The Prehistory of the Albanian Vowel System" (2008) is the most recent specialist treatment of the relevant phonology. Three passages are decisive.

The unresolved source of length. On p. 592, Vermeer lists candidate sources of Albanian vowel length and places *dorë* in a category characterized by the unresolved contest between Demiraj/Orel's compensatory-lengthening account and Huld's analogical-lengthened-grade account: "Although the jury is still out on (3) and (4), nothing in what follows depends on a decision about these cases." Within the standard view itself, the source of length in *dorë* is unresolved.

The reconstruction without sibilant. On p. 595, Vermeer gives his own reconstruction: "**ē* > *o* (= **ā*), e.g. *dorë* 'hand' < **ĝh^hēr-* (the source of the length of **ē* is controversial in this example, see section 2)." The absence of *-s-* is significant. Vermeer writes **ĝh^hēr-*, not **ĝh^hesr-*. He has implicitly accepted Belardi's and Pokorny's argument that **ĝh^hēs^r-* would have given Albanian **dostrë*, and has reconstructed a sibilant-less form to escape the problem. Vermeer's **ĝh^hēr-* is structurally identical to the **der*-side **dēr-* except for the initial consonant.

The unreliable sound law. Vermeer's footnote 3 on the **ō* > *e* reflex adds a further point. Discussing the sound law on which Huld's dismissal of La Piana's **dōrom* depended, Vermeer writes: "Despite complete consensus about this reflex I do not think there is a single example all interested scholars regard as reliable." The very sound law that filtered out La Piana's reconstruction is described by Vermeer as having no reliably agreed example.

7. The Reconstruction

Albanian *dorë* descends from a feminine **-eh₂-* derivative built on the Szemerényi'd nominative singular of a consonant-stem r-stem:

$$*der-s \rightarrow *dēr \rightarrow *dēr-eh_2- \rightarrow *dēr-ā \rightarrow dorë$$

Each step uses sound-law rules that are well-established in independent IE morphology.

Step 1: **der-s*. The e-grade r-stem with nominative singular ending **-s*. Standard PIE consonant-stem inflection.

Step 2: **dēr* (Szemerényi's Law). Szemerényi's Law: pre-PIE **-VRs#* → PIE **-VR#* (Szemerényi 1996: 113–119; Lundquist and Yates 2018: 2083). Standard examples: **ph₂tēr* “father” (← **ph₂tér-s*), **kwōn* “dog” (← **kwén-s*). Applied to **der-s*, the Law produces **dēr* by exactly the same mechanism. The long **ē* is a phonological consequence of a standard sound rule in its standard environment.

Step 3: **dēr-eh₂-*. The Szemerényi'd nominative **dēr* is consonant-final and animate-gendered by default. The standard morphological process for feminine reanalysis is the **-eh₂-* extension: a consonant-stem nominative is reanalyzed as the base for a derived feminine **-eh₂-* stem. This is the same morphological move Hamp 1966 invoked for **ġhē(s)rā* and Huld 1984 adopted. If the pattern is acceptable for the **ġh^hesr*-side, it is acceptable for the **der*-side.

Step 4: **dēr-ā*. Standard post-PIE laryngeal loss: **-eh₂ → *-ā*.

Step 5: *dorë*. Regular Albanian **ē → o* (Pedersen §34; Tagliavini 1936: 14; Huld 1984: 54; Vermeer 2008). Unstressed final **-ā → -ë*.

The proto-Albanian form **dērā / *dērā* arrived at by this chain is identical to Demiraj's own reconstruction for proto-Albanian (AE p. 140), reached on the **ġh^hesr*-side through a chain that requires the **ġh^h → d* shift. The **der*-side reading reaches the same proto-Albanian form from PIE **der-s* using only regular sound changes.

The plural. The consonantal plural *duar* with u-vocalism is irregular and is not directly accounted for by the **dēr-eh₂-* singular reconstruction. This irregularity is shared across the published candidates; no current reconstruction gives a fully transparent account of both *dorë* and *duar*. The reading developed here restricts its claim to the singular.

Parity with the **ġh^hesr*-side. The **-eh₂-* feminine extension on a Szemerényi'd consonant-stem base is not a frequently attested PIE category outside the contested **ġh^hēsra* family of reconstructions. The morphological pattern is the same for both reconstructions; if it is acceptable for **ġh^hēsra* it is acceptable for **dēr-eh₂-*. The **der*-side reading wins on phonological parsimony: regular **d → d* versus irregular **ġh^h → d*. And it wins on phonological grounding: Szemerényi's Law for the lengthening, against three competing accounts for the length on the **ġh^hesr*-side.

**dër-eh*₂- uses only the sound laws all parties accept.

8. The Deeper Question: Where Does PIE **de-r-* Come From?

A reconstruction of Albanian *dorë* from PIE **de-r-* “hand, open palm” leaves a further question: where does **de-r-* itself come from? The IEW 203 entry presents **der-* as a primary root without a deeper derivation. But hand-words across the world’s languages frequently arise from a small set of source domains, and the typological literature documents the relevant pathways with some precision. This section develops the case for the ARM → HAND hypothesis and states the formal constraint that qualifies it.

8.1 Heine and Kuteva on ARM → HAND

Heine and Kuteva’s *World Lexicon of Grammaticalization* (2002) documents the cross-linguistic pathways by which body-part terms shift in scope between adjacent parts of the body. ARM → HAND is among the well-attested channels: a lexeme whose original reference was the upper limb acquires a hand reading through the metonymy of the distal portion. Languages with a single lexeme covering both “arm” and “hand” are widespread — Russian *ruka*, Polish *ręka*, Czech *ruka*, Hebrew *yad* — and where two distinct lexemes coexist, one is often a derivative or specialization of the other. A hand-noun whose deeper origin is not directly recoverable from the comparative evidence may plausibly descend from an arm-noun, and this scenario is preferred when the formal evidence is consistent with it.

8.2 Matasóvič 2009: A Published Indo-European Parallel

The ARM → HAND pathway is registered within Indo-European etymology in the published *Etymological Dictionary of Proto-Celtic* (Matasóvič 2009: 103–104). Matasóvič’s entry for Proto-Celtic **dowsant-* reconstructs it as a compound of the PIE arm-root **dows-* “arm, fore-arm” (IEW 226) with a second element **-ant-* “border, limit” (cf. Gothic *andeis* “end”), proposing the compositional semantics “arm-limit”, “the end of the arm” — with the hand-meaning emerging from the arm-meaning through this elaboration. The published Old Irish forms *doë*, *doe* are glossed as both “arm” and “hand”; the same lexeme straddles the two readings within attested Celtic. The ARM → HAND pathway is thus documented within Indo-European etymology by a standard reference work, with the morphological mechanism made explicit.

8.3 The Sanskrit Evidence: *doṣ-/dor-* and the Metrology Cluster

The Sanskrit s-stem arm-noun *doṣ-/doṣán-* is the principal IE witness for **dóws-*, attested from the Rigveda forward. Böhtlingk-Roth notes that *doṣan-* “occurs only in nom. dual *doṣañī*” in the Rigveda and Atharvaveda — morphologically natural for an arm-noun, since arms come in pairs. The compound stem *dor-*, produced by regular

Sanskrit visarga sandhi before voiced segments, yields a rich and productive system attested from Vedic prose forward (Mayrhofer EWAia I.749):

- *dor-bāhū* “the arms” (Śatapatha-Brāhmaṇa)
- *dor-mūla* “root of the arm” = shoulder
- *dor-daṇḍa* “arm-stick” (epithet of Arjuna Kārtavīrya)
- *dor-nikartana* “sword” (literally “arm-cutter”)
- *dor-jyā* “arm-string” = sine (Sūryasiddhānta 3.10)
- *doḥsahasrabhṛt* “thousand-armed” (Mahābhārata)

The metrological readings are decisive for the framework: *dos* as a unit of 18 inches in the *Mātaṅgalīlā* 1.14 (a Sanskrit cubit-class measurement), and *dor-jyā* “arm-string” = sine in the *Sūryasiddhānta* 3.10. The Sanskrit mathematical tradition built its trigonometry on the arm-as-measurement metaphor; the *Sūryasiddhānta* preserves the body-part anchor in its mathematical vocabulary. The cross-branch parallel is the Indo-European cubit family: Latin *cubitus* “elbow; cubit”; Greek πῆχυς “forearm; cubit”; Old English *ell*; Old High German *elina* (whence *elbow* < OE *eln-boga* “ell-bow”). The Sanskrit *dos* = 18 inches sits squarely inside this cross-IE pattern.

The ARM → HAND shift within Sanskrit itself. The *Śivapūraṇa* 2.4.18 has Pārvatī “taking up her son with both her hands (*dos*)” — the arm-word used in late Sanskrit to mean “hands.” This is not a comparative observation across IE branches but a within-Sanskrit attestation: the same lexeme that means “arm, fore-arm” in Vedic prose has shifted to “hand” by the Purāṇic literature, through the standard mechanism of distal-part metonymy.

The cross-branch cognate set. The s-stem arm-noun **dóws-* is attested across the IE family: Avestan *daoš-* “upper arm, shoulder”; Pahlavi *dōš* “shoulder, arm”; Khotanese *durra-*; Latvian *pa-duse* “armpit” (literally “by-arm”), from Proto-Balto-Slavic **dus-*, the zero grade of **dóws-* (Karulis); Old Church Slavonic *pazuxa* “bosom” — the same **pa-* compound with the semantic chain “by-arm” → “armpit” → “bosom”; Proto-Celtic **dowsant-* (Old Irish *doë, doe*) on the Matasóvič reading of §8.2.

8.4 The Morphological Frame: the r/n Heteroclite Class and Locative *-r

The morphological apparatus on which the framework’s **de-r-* depends is supplied by three convergent published treatments of the PIE **-r* element.

Lipp 2009 on the body-part heteroclite class. Lipp’s *Die indogermanischen und einzelsprachlichen Palatale im Indoiranischen* (Heidelberg: Winter; the *Thorn-Problem* volume) develops the formal apparatus for the archaic PIE r/n heteroclite paradigm — the class of nouns whose strong stem ends in **-r* and whose weak stem ends in **-n*, with the alternation case-conditioned (nominative-accusative

on the *r*-stem, oblique cases on the *n*-stem). Standard exemplars: **péh₂wṛ* “fire” (Greek *nῦρ*, Hittite *paḥḥur*) / **ph₂un-* (oblique); Sanskrit *yákṛt* “liver” / *yaknás* (oblique). Lipp supplies the formal apparatus for the body-part heteroclite subclass; the framework’s **de-r-* belongs to this class.

Bauhaus 2019 on PIE **-r* as a locative case-ending. Felix Bauhaus’s chapter in Kloekhorst and Pronk (2019: 15–29) argues that PIE **-r* was a real locative case-ending, not an “adverbial ending.” The pronominal evidence is decisive: **k^wor* “where,” **tor* “there,” Sanskrit *kár-hi* / *tár-hi*, Gothic *ƿar* / *þar* / *hēr*, Latin *cūr* < **quōr*, Lithuanian *kuř*, Latvian *tūr*, Armenian *ur*, Albanian *kur*. On pronouns, the **-r* is transparently a case-ending. Albanian *kur* “when” supplies an Albanian-internal attestation of the locative **-r* case-ending — in the same Albanian lexicon that preserves *dorë*.

Kahl 2024 on the locative origin of **-ero-*. Kahl’s Harvard dissertation *Diachronic Studies in Indo-European Degree Morphology* contains a section (§2.3.1) titled “The locative origin of **-ero-*,” tracing the **(t)ero-* comparative suffix to an archaic locative morpheme **-er*. Kahl independently arrives at the same conclusion as Bauhaus from a third direction. The three treatments — Lipp from heteroclitic nominal morphology, Bauhaus from pronominal case-marking, Kahl from degree morphology — are methodologically independent; their convergence on the locative status of PIE **-r* supplies the morphological foundation for **de-r-*.

8.5 The Formal Constraint: e-grade **de-r-* vs. o/u-grade **dóws-*

A formal constraint qualifies the deeper hypothesis. The well-attested IE arm-root is **dóws-/*dóus-* with o/u-vocalism throughout. Matasóvič’s **dowsant-* is built on **dows-*, not on anything resembling **der-*. The Sanskrit, Avestan, Khotanese, and Balto-Slavic forms all preserve o/u-vocalism. The cleanest cross-branch ARM → HAND example (PCelt. **dowsant-*) therefore sits on a formally distinct root from the one needed for *dorë*. The vocalism does not match.

This means the deeper hypothesis cannot run through the surface route: PIE **de-r-* “hand” is not simply the e-grade form of the arm-noun **dóws-*, because the arm-noun’s vocalism is consistently o/u, not e. If **de-r-* is to be related to the arm-noun at all, the relation must be at a deeper Pre-PIE level — at which e-grade and o/u-grade forms might be derivable from a common compositional base — rather than at the synchronic PIE level. The broader four-paper project develops that Pre-PIE compositional analysis; the present paper does not undertake it.

8.6 Implications: A Qualified ARM → HAND Reading

The hypothesis that PIE **de-r-* itself derives from an arm-related lexeme by the ARM → HAND pathway is therefore qualified rather than absolute. What is well-supported: the cross-linguistic typological pathway (Heine and Kuteva 2002); the published Indo-European parallel in Matasóvič’s **dowsant-* entry; the Sanskrit

s-stem arm-noun *doṣ-/dor-* with its rich compound system, metrology, and within-Sanskrit ARM → HAND attestation; the morphological frame of Lipp 2009, Bauhaus 2019, and Kahl 2024; and the cross-branch arm-noun cognate set (Avestan, Khotanese, Balto-Slavic, Celtic). What is constrained: the well-attested IE arm-root has o/u-vocalism and the cleanest ARM → HAND example sits on a formally distinct root, so the deeper unification requires a Pre-PIE compositional analysis developed in the companion papers.

The deeper hypothesis is not necessary for the etymology of *dorë* developed in §7. The phonological derivation *dorë* < **dēr-eh₂-* < **dēr* < **der-s* stands regardless of where **de-r-* itself comes from.

9. Conclusion

The case for the **der*-side reading of Albanian *dorë* rests on three convergent grounds.

Phonology. **de-r-* → *dorë* uses only sound laws all parties accept: Szemerényi's Law on the e-grade r-stem nominative **der-s* yielding **dēr*; the standard **-eh₂-* feminine extension yielding **dēr-eh₂-*; post-PIE laryngeal loss yielding **dēr-ā*; regular Albanian **ē* → *o* and unstressed final **-ā* → *-ë* yielding *dorë*. Every **ġh^hesr*-side reconstruction must accommodate a step **ġh^h* → *d* in Albanian, where the regular reflex of PIE **ġh^h* is *dh*.

Historiography. The 1990 consolidation of the **ġh^hesr*-side reading rests, in its foundational citation, on a page reference (Tagliavini AD 101) whose content does not address the question. Pokorny's own text at IEW 447 had stated, in his own voice, that **ġh^hesr-* could not produce *dorë* — supplying the Belardi argument in the very page Demiraj cites as the position to be overturned. The citation that grounds the consolidation does not, on inspection, deliver the rejection the consolidation requires.

Convergence. The post-2000 specialist treatment (Vermeer 2008) has independently moved to **ġh^hēr-* — a sibilant-less form differing from **dēr-* by one phoneme. Vermeer describes the source of length in *dorë* as unresolved within the **ġh^hesr*-side framework, and describes the sound law on which Huld's dismissal of La Piana's **dōrom* rested as having no reliably agreed example. Taken on its own internal terms, the standard reading is unsettled.

The deeper hypothesis that PIE **de-r-* itself derives from an arm-related lexeme by the ARM → HAND pathway receives qualified support from cross-linguistic typology (Heine and Kuteva 2002), from the Matasóvič Proto-Celtic parallel (2009: 103–104), from the Sanskrit *doṣ-/dor-* evidence with its within-Sanskrit ARM → HAND attestation, and from the morphological frame of Lipp 2009, Bauhaus 2019, and Kahl 2024. The formal constraint — the well-attested IE arm-root is **dóws-* (o/u-grade), not **der-* (e-grade) — means the deeper unification requires a Pre-PIE

compositional analysis developed in the companion papers. The deeper hypothesis is speculative and does not bear the same evidentiary load as the §7 reconstruction.

Probabilities. Conditional on the **de-r- → dorë* etymology being correct: 55–70%. The reading uses only regular sound laws, finds the post-2000 specialist reconstruction one phoneme away, and rests on a published lineage running from Meyer 1883 through Pokorny 1959 to Frisk and the DGE. The chief residue is the irregular plural *duar*, which no current reconstruction handles transparently. Conditional on the further hypothesis that PIE **de-r-* itself derives from an earlier arm-word by the ARM → HAND pathway: 15–25%. The pathway is cross-linguistically robust and has a published Indo-European precedent in **dowsant-*, but the formal constraint (e-grade vs. o/u-grade) shows that the deeper unification requires a Pre-PIE compositional analysis not undertaken here.

Authorship and Contributions

This paper is co-authored by J.M. Chavis (human) and Claude (Anthropic AI; primarily Claude Opus 4.7).

Chavis’s contributions. Chavis articulated the proposal to develop the *dorë* argument as a standalone paper; directed the editorial decisions on cognate restraint; shaped the framing of the historiographic discussion to treat the post-1990 consolidation as scholarly inheritance rather than individual error; directed the §8 expansion to integrate the Sanskrit *doṣ-/dor-* evidence and the K&P Bauhaus material on PIE **-r-*; and determined the scope at each revision stage. All fundamental decisions about framing and direction rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude’s contributions. Claude carried out the literature reading, citation verification (Pokorny IEW pp. 203, 226, 447; Demiraj AE p. 140; Tagliavini AD pp. 101, 106; Tagliavini’s 1936 inaugural lecture pp. 13–14; Hamp 1966 p. 115; Huld 1984 p. 54; Vermeer 2008 pp. 592–595; Matasóvič 2009 pp. 103–104; Mayrhofer EWAia I.749; the Sanskrit lexicographical sources for *dor-*; Bauhaus 2019 in Kloekhorst & Pronk 2019; Lipp 2009; Kahl 2024), and technical drafting under Chavis’s direction. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management.

Both authors take responsibility for the framework and the specific claims made in this paper.

Select References

Bauhaus, F. 2019. “The Proto-Indo-European Suffix **-r* Revisited.” In Kloekhorst, A. and T. Pronk (eds.), *The Precursors of Proto-Indo-European*, 15–29. Leiden: Brill.

- Beekes, R. 2010. *Etymological Dictionary of Greek*. Leiden: Brill.
- Belardi, W. 1948. "Ittito *ki-eš-sar*, greco χείρ." *Rivista degli Studi Orientali* 23: 69–81.
- Böhtlingk, O. and R. Roth. 1855–1875. *Sanskrit-Wörterbuch*. St. Petersburg: Kaiserliche Akademie der Wissenschaften.
- Bopp, F. 1854. *Über das Albanesische in seinen verwandtschaftlichen Beziehungen*. Berlin: Akademie der Wissenschaften.
- Chavis, J.M. and Claude. 2026. *The Hand That Gives: Pre-PIE *de-, the *de-r- hand-and-palm cluster, and the etymology of Greek δῶρον 2 "palm."* Companion paper.
- Demiraj, B. 1990. "Sull'etimologia di alb. *dorë*." *Indogermanische Forschungen* 95: 99–110.
- Demiraj, B. 1997. *Albanische Etymologien*. Amsterdam: Rodopi.
- Diccionario Griego-Español* (DGE). <https://dge.cchs.csic.es>
- Frisk, H. 1960–72. *Griechisches Etymologisches Wörterbuch*. Heidelberg: Winter.
- Hamp, E.P. 1966. "The Position of Albanian." In Birnbaum, H. and J. Puhvel (eds.), *Ancient Indo-European Dialects*, 97–121. Berkeley: University of California Press.
- Heine, B. and T. Kuteva. 2002. *World Lexicon of Grammaticalization*. Cambridge: Cambridge University Press.
- Huld, M.E. 1984. *Basic Albanian Etymologies*. Columbus, OH: Slavica.
- Kahl, L. 2024. *Diachronic Studies in Indo-European Degree Morphology*. PhD dissertation, Harvard University.
- Karulis, K. 1992. *Latviešu etimoloģijas vārdnīca*. Rīga: Avots.
- Kloekhorst, A. and T. Pronk (eds.). 2019. *The Precursors of Proto-Indo-European*. Leiden: Brill.
- La Piana, M. 1937. *Intorno al riflesso della vocale "o" lunga dall'indoeuropeo e dal latino nell'albanese*. Palermo.
- La Piana, M. 1942. [Note on Albanian *dorë*]. *Indogermanische Forschungen* 58: 98.
- Lipp, R. 2009. *Die indogermanischen und einzelsprachlichen Palatale im Indoiranischen*. 2 vols. Heidelberg: Winter.
- Lundquist, J. and A.D. Yates. 2018. "The Morphology of Proto-Indo-European." In Klein, J. et al. (eds.), *Handbook of Comparative and Historical Indo-European Linguistics*, vol. 3, 2079–2195. Berlin: De Gruyter.
- Matasóvič, R. 2009. *Etymological Dictionary of Proto-Celtic*. Leiden: Brill.
- Mayrhofer, M. 1992–96. *Etymologisches Wörterbuch des Altindoarischen* (EWAia). Heidelberg: Winter.
- Meyer, G. 1883. *Albanesische Studien I*. Vienna: Karl Gerold's Sohn.
- Meyer, G. 1891. *Etymologisches Wörterbuch der albanesischen Sprache*. Strassburg: Trübner.
- Orel, V. 1998. *Albanian Etymological Dictionary*. Leiden: Brill.
- Pedersen, H. 1909, 1913. *Vergleichende Grammatik der keltischen Sprachen*. Göttingen: Vandenhoeck & Ruprecht.
- Pokorny, J. 1959. *Indogermanisches etymologisches Wörterbuch* (IEW). Bern: Francke.
- Schindler, J. 1967. "Tocharische Miscellen." *Indogermanische Forschungen* 72: 239–249.
- Stifter, D. 2022. "Continental Celtic DERNV[at the Bronze of Novallas." *Études Celtiques* 48: 41–60.

- Szemerényi, O.J.L. 1996. *Introduction to Indo-European Linguistics*. Oxford: Oxford University Press.
- Tagliavini, C. 1936. *La lingua albanese*. Inaugural lecture, University of Padua. *Studi Albanesi* V-VI [1935-36]: 5-33.
- Tagliavini, C. 1937. *L'albanese di Dalmazia: contributi alla conoscenza del dialetto ghego di Borgo Erizzo presso Zara*. Florence: Olschki.
- Vermeer, W. 2008. "The Prehistory of the Albanian Vowel System: A Preliminary Exploration." In Lubotsky, A. et al. (eds.), *Evidence and Counter-Evidence*. Amsterdam: Rodopi, vol. 1: 591-608.